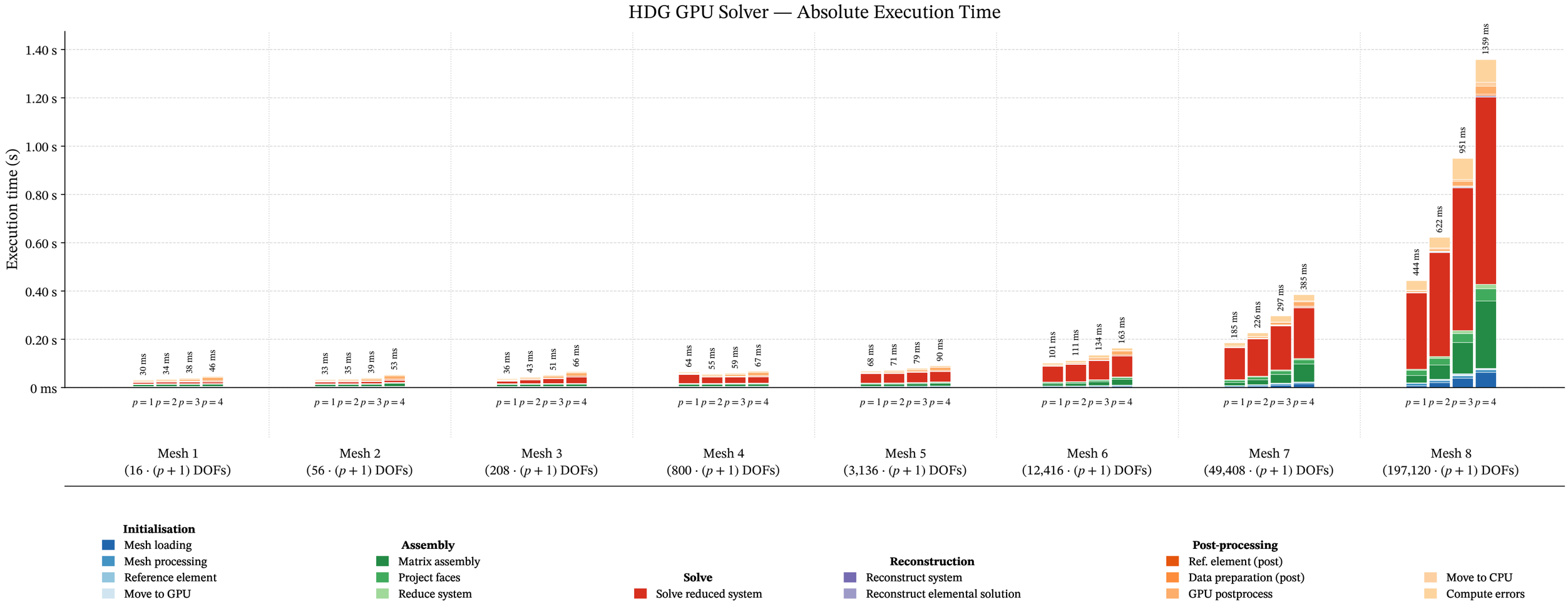
HDG GPU Solver — Absolute Execution Time
Execution time (s)
1.40 s
1.20 s
1.00 s
0.80 s
0.60 s
0.40 s
0.20 s
0 ms
30 ms
34 ms
38 ms
46 ms
33 ms
35 ms
39 ms
53 ms
36 ms
43 ms
51 ms
66 ms
64 ms
55 ms
59 ms
67 ms
68 ms
71 ms
79 ms
90 ms
101 ms
111 ms
134 ms
163 ms
185 ms
226 ms
297 ms
385 ms
444 ms
622 ms
951 ms
1359 ms
p = 1 p = 2 p = 3 p = 4
Mesh 1 (16 · (p + 1) DOFs)
Mesh 2 (56 · (p + 1) DOFs)
Mesh 3 (208 · (p + 1) DOFs)
Mesh 4 (800 · (p + 1) DOFs)
Mesh 5 (3,136 · (p + 1) DOFs)
Mesh 6 (12,416 · (p + 1) DOFs)
Mesh 7 (49,408 · (p + 1) DOFs)
Mesh 8 (197,120 · (p + 1) DOFs)
Initialisation
Mesh loading
Mesh processing
Reference element
Move to GPU
Assembly
Matrix assembly
Project faces
Reduce system
Solve
Solve reduced system
Reconstruction
Reconstruct system
Reconstruct elemental solution
Post-processing
Ref. element (post)
Data preparation (post)
GPU postprocess
Move to CPU
Compute errors